"CAN WE TAX CARBON AT A SOCIALLY OPTIMAL LEVEL?"

Group 6

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METHODOLOGY & EMPIRICAL STRATEGY

General Optimal Control Formulation We define our variables as follows:

Variable	Symbol	Remark
Output	Y_t	> 0
Abatement/efficiency capital (state)	k_t	-
Carbon stock or "damage driver" (state)	S_t	-
Control (abatement investment)	i_t	-
Emissions intensity	$\mu(k_t)$	$\mu'(k) < 0$
Emissions flow	E_t	$E_t = \mu(k_t)Y_t$
Damages	$\mathcal{D}(S_t)$	$D'(S) > 0,\ D''(S) \geq 0$
Abatement investment cost	$\mathcal{C}(i_t)$	$C'(i) > 0,\ C''(i) > 0$
Depreciation of k	δ	> 0
Carbon decay	ϕ	> 0
Social discount rate	ρ	> 0

Table 1: Variables & System States

METHODOLOGY & EMPIRICAL STRATEGY

The empirical strategy is an econometric analysis... but what model will we use?

One possibility is a mixed log-linear model with carbon tax as one of the regressors:

$$\ln E_{it} = \alpha_i + \theta \ln Y_{it} - \eta \, \mathtt{CarbTax}_t + \varepsilon_{it}$$

More research will tell us which functional form is better, but a general form is simply:

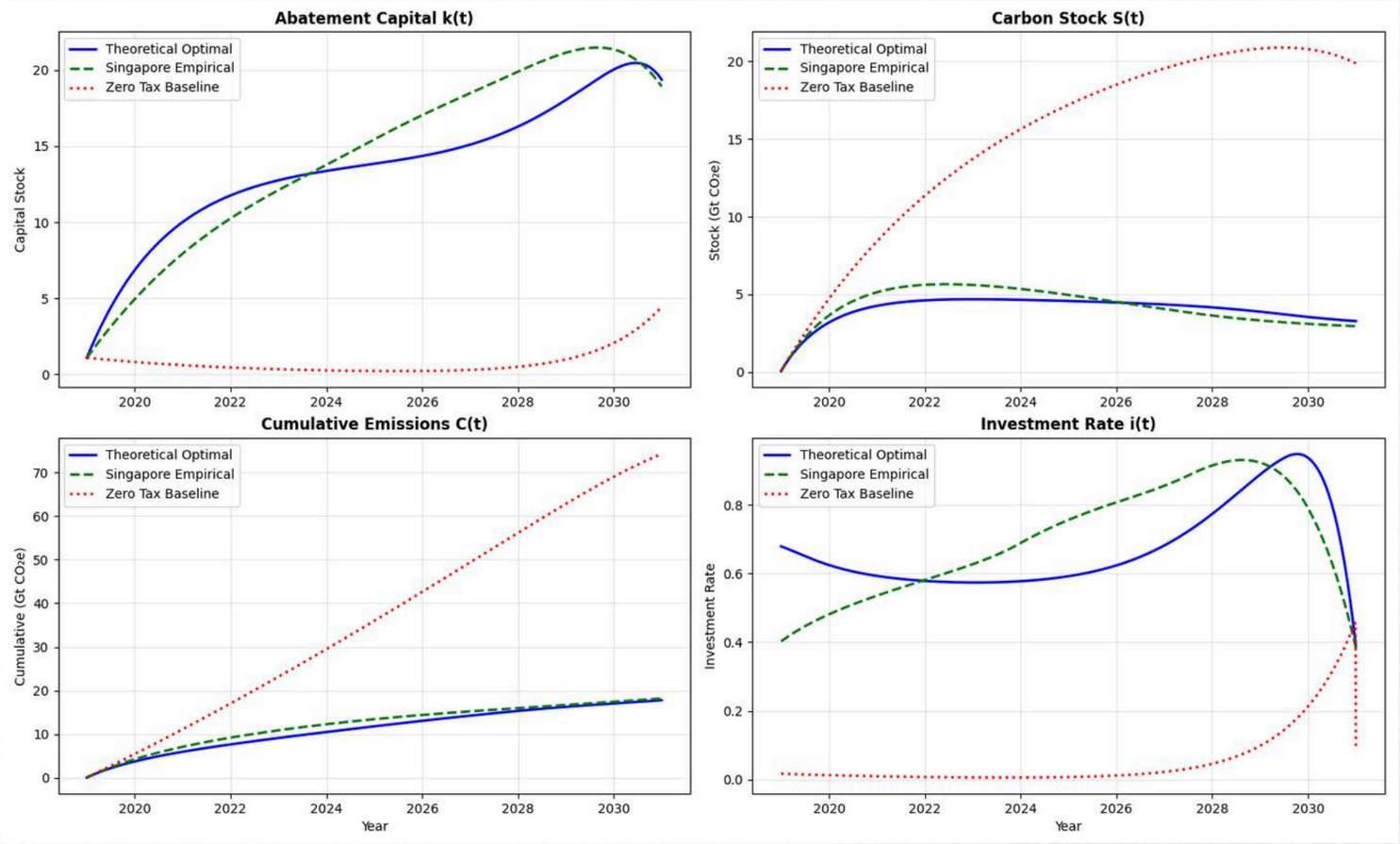
$$\mathcal{L}(y_{it}) = \Phi(\mathbf{x}_{it}, \boldsymbol{\beta}_i) + \varepsilon_{it}$$

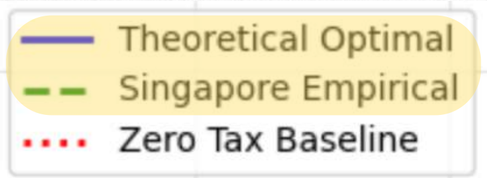
$$f(y_{it}) = \Phi(\mathbf{x}_{it}, \boldsymbol{\beta}_i) + \varepsilon_{it}$$

We compare & contrast theoretical simulations with empirical (econometric analysis) to identify the so-called optimality wedge - what can or need to do to achieve the desired steady state!

FINDINGS

Current Singapore Policy vs. Theoretical Optimal

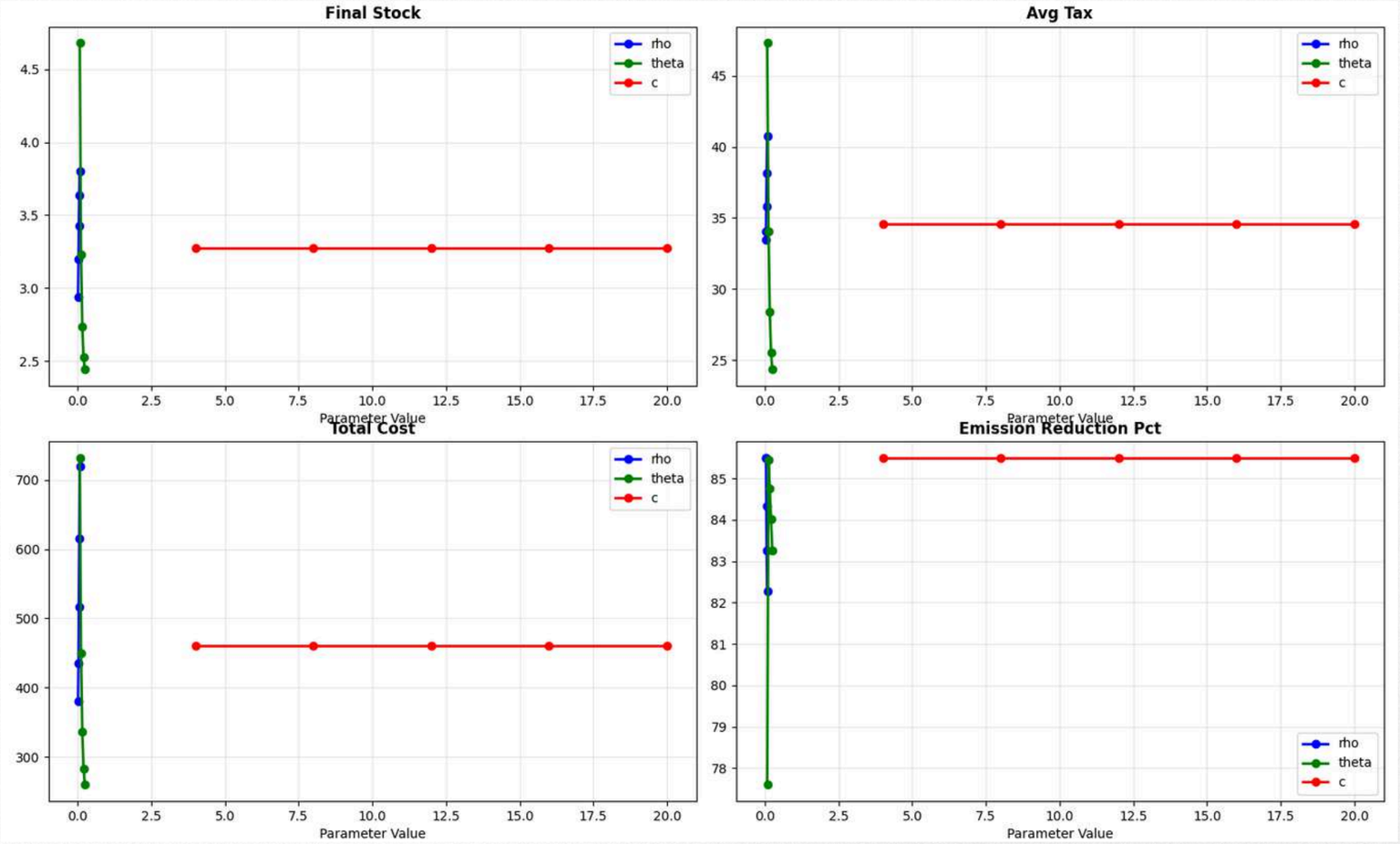




- Abatement capital - SG spending more than is efficient, same direction
- Carbon Stock - SG slightly underperforming
- Cumulative Emissions - SG slightly underperforming but same trajectory
- Investment rate - SG spends more in earlier years

FINDINGS

Sensitivity Analysis, in One Year

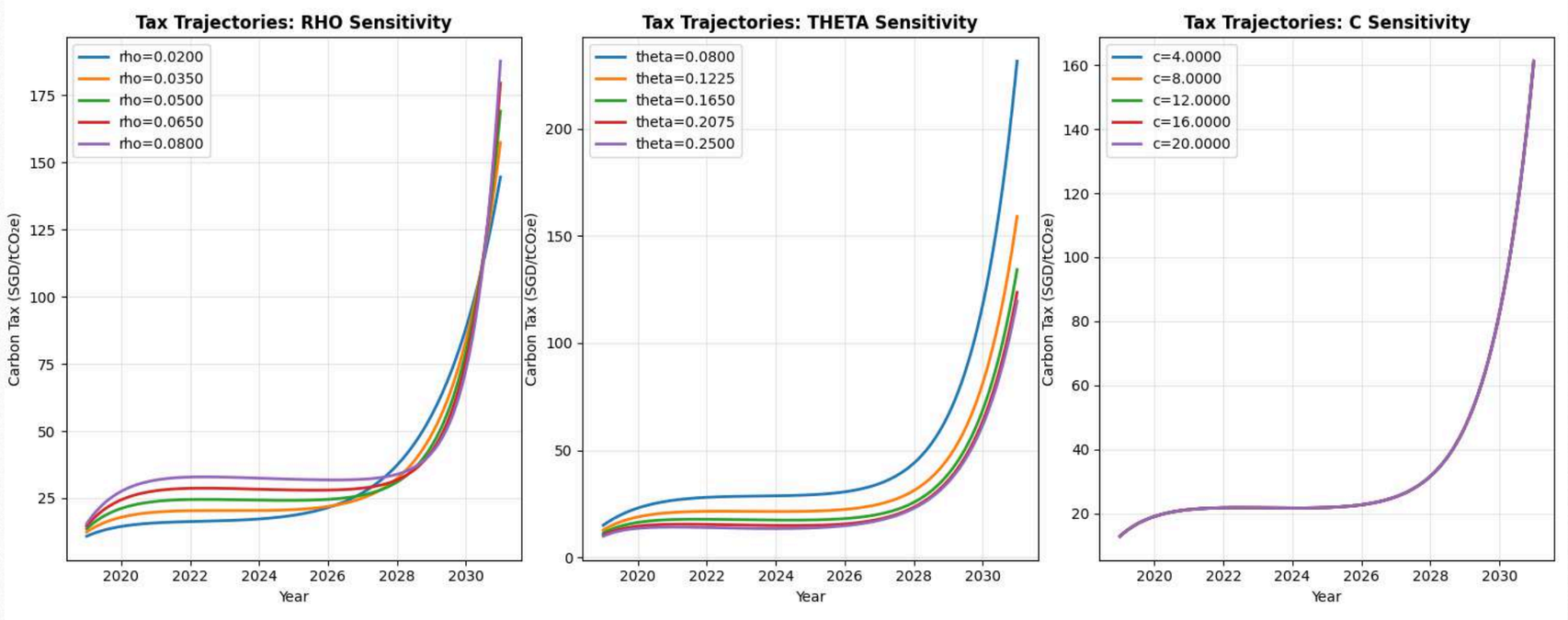


- rho = social discount rate
- theta = carbon decay
- c = abatement cost

effects on Capital Stock, Tax, Total Cost and Emission Reduction Percentage

FINDINGS

Sensitivity Analysis, Trajectory in 10 Years

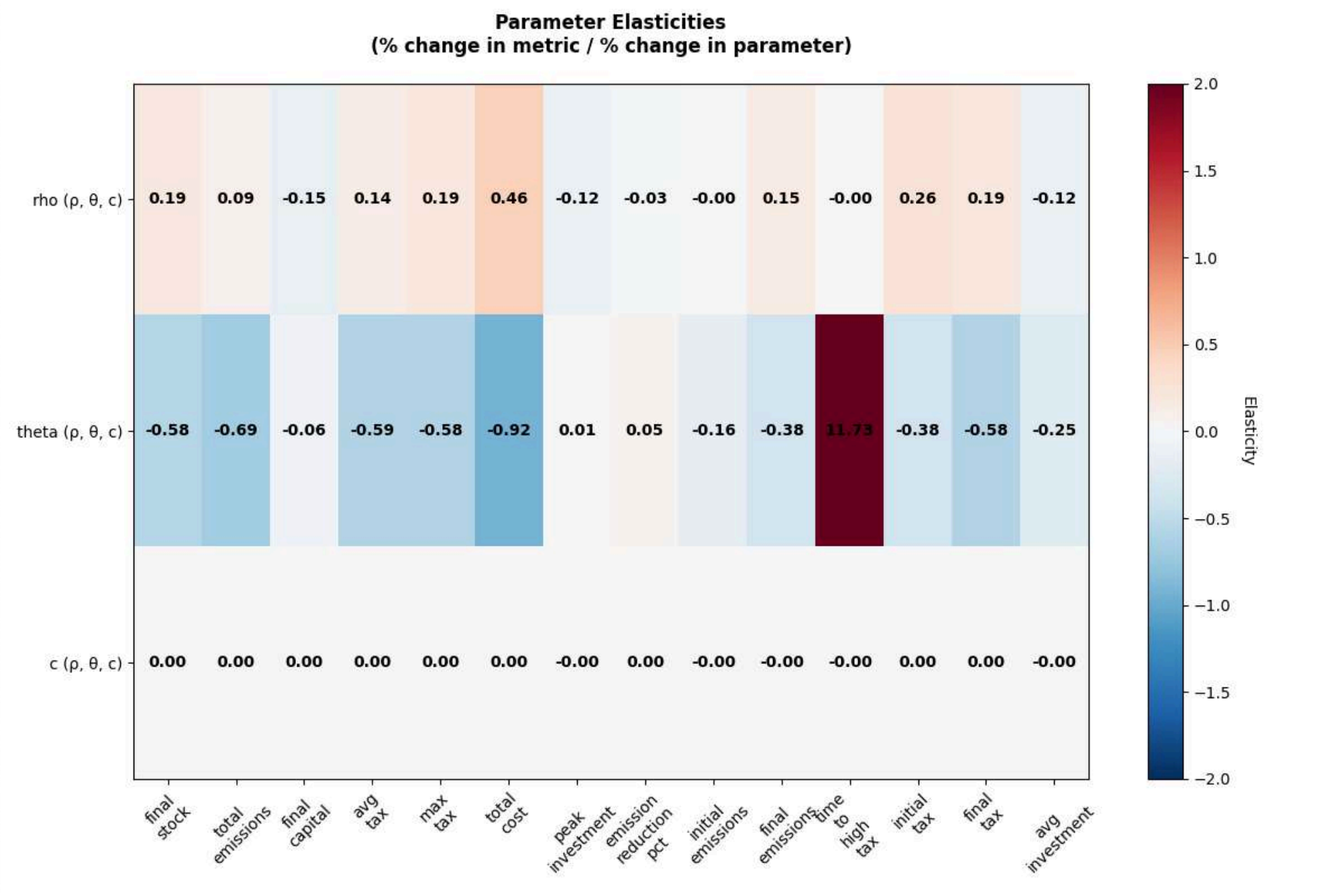


- rho = social discount rate
- theta = carbon decay
- c = abatement cost

each graph demonstrate how the carbon tax will differ in 10-year-time with variation of each parameter
(rho, theta, and c). rho and theta has a significant effect on carbon tax.

FINDINGS

Parameter Elasticity



- In general, our 4D Hamiltonian system is almost fully insensitive to abatement costs (ie the trajectory of taxes is not affected by abatement cost)
- Social discount rate is relatively inelastic while emissions intensity (carbon decay) sensitivity is more elastic

- Note: This is dependent on how the model is defined, which directly impacts the state dynamics of the system

SUMMARY OF FINDINGS

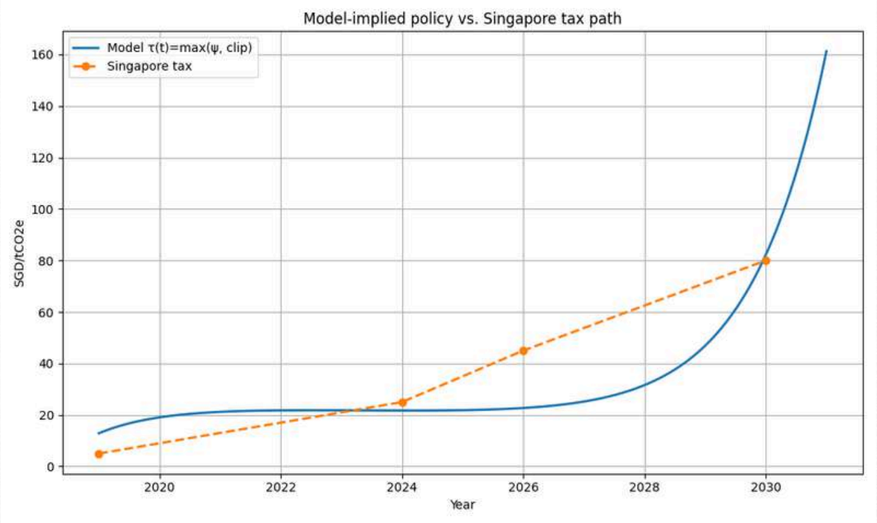
Empirical trajectory reducing carbon emissions at a similar rate as is **theoretical** counterpart, but is spending and investing more than efficient level

Abatement cost does not effect our tax trajectories or predictions of carbon emissions, but as emissions intensity increases, the tax trajectories increase. As social discount increases, tax trajectories also increase. Carbon emissions vary about 7% from changes in emissions intensity, and about 3% from social discounts.

Social discount rate is relatively inelastic, leading to an smaller change in taxes as it changes, while emissions intensity is more elastic, leading to a greater change in taxes as it changes

IMPLICATIONS: VALUE ADDED

Long-run tax rate/trajectory is indeed socially optimal - both models intersect at 2030. Hence, in the long-run, the current tax model is efficient. However, the divergence between the 2 curves shows us what changes can be made to improve this policy in its year-to-year course, and become even more socially optimal



We use a well-established theoretical approach (Hamiltonian optimal control theory) to analyse Singapore's carbon tax policy, with no precedent in Singapore-focused environmental economics research.

Policy recommendations that are supported by our model to help increase dynamic efficiency and social optimal welfare.

POLICY IMPLICATIONS

Increasing abatement capital in the earlier years, and reducing carbon taxes in the later years - having an adaptable tax model that is more sensitive to estimates of the marginal social damage per tonne of CO_2 (SCC)

The need for transitional equity and and a tax-and-trade policy

The need for international and regional coordination and a convincing tax model for use in these efforts

CHALLENGES IN PROJECT

Utility function - logarithmic; damage function - quadratic

- Assumed for simplicity
- May not be true in real life
- Should be matched with collected data

Carbon tax is uniform, across sectors, at all times

Data Reliability
Historical data bias → Overfitting

In real life, dynamical system is almost surely even higher dimensional

THANK YOU